



Applied Data Mining

**— Week 5-1—
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Outline

- Classification: Basic Concept



Classification: Basic Concept

- It is a form of data analysis that extracts models describing important data classes.
- Such models, called classifiers, predict categorical (discrete, unordered) class labels.
- For example, we can build a classification model to categorize bank loan applications as either safe or risky.
- Such analysis can help provide us with a better understanding of the data at large.

Supervised Vs Un-Supervised Learning

- **Supervised learning (classification)**
 - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
 - New data is classified based on the model trained on a training set
- **Unsupervised learning (clustering)**
 - The class labels of training data is unknown
 - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Feature	Supervised	Unsupervised
Labels	Available	Not available
Goal	Predict output	Discover structure
Example	Classification	Clustering

Prediction Problems: Classification vs. Numeric Prediction

■ Classification

- Predicts categorical class labels (discrete or nominal)
- Classifies data (constructs a model) based on the training set and the values (**class labels**) in a classifying attribute and uses it in classifying new data

■ Numeric Prediction

- Models continuous-valued functions, i.e., predicts unknown or missing values

■ Typical applications

- Credit/loan approval
- Medical diagnosis: if a tumor is cancerous or benign
- Fraud detection: if a transaction is fraudulent
- Web page categorization: which category it is

Classification—A Two-Step Process

- **Model construction**: describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- **Model usage**: for classifying future or unknown objects
 - **Estimate accuracy** of the model
 - The known label of test sample is compared with the classified result from the model
 - **Accuracy** rate is the percentage of test set samples that are correctly classified by the model
 - **Test set** is independent of training set (otherwise over fitting)
 - If the accuracy is acceptable, use the model to **classify new data**

Process (1): Model Construction

